

Q1: Multiple Choice (20 pts)

Choose the correct answer in the following:

1. The minimum element in the Binary Search Tree (BST) is located at \_\_\_\_?
  - a. The rightmost node of the tree
  - b. the leftmost node of the tree
  - c. The leftmost node of the left subtree for the root
  - ☒ d. b+c
2. To check if a node in a BST is a leaf
  - a.  $\text{if}(\text{node} \rightarrow \text{left} == \text{NULL} \parallel \text{node} \rightarrow \text{right} == \text{NULL})$
  - ☒ b.  $\text{if}(\text{node} \rightarrow \text{left} == \text{NULL} \&\& \text{node} \rightarrow \text{right} == \text{NULL})$
  - c.  $\text{if}(\text{node} \rightarrow \text{left} != \text{NULL} \&\& \text{node} \rightarrow \text{right} != \text{NULL})$
  - d.  $\text{if}(\text{node} \rightarrow \text{left} != \text{NULL} \parallel \text{node} \rightarrow \text{right} != \text{NULL})$
3. A(n) \_\_\_\_\_ is a list of homogenous elements in which the addition and deletion of elements occurs only at one end.
  - ☒ a. Stack
  - b. Queue
  - c. array
  - d. linked list
4. Except for the first and last node, every node in a doubly linked list contains \_\_\_\_\_.
  - a. No address information
  - b. The address of the next node
  - ☒ c. the address of the previous node
  - d. b and c
5. Suppose the first Fibonacci number is 0 and the second is 1. What is the sixth Fibonacci number?
  - a. 5
  - b. 6
  - c. 7
  - ☒ d. 8

8.9
6. The worst case for binary search occurred when ...?
  - a- The key is in the middle of the array
  - b- The key is in the first position of the array
  - ☒ c- The key is at the last position of the array

Q4: Trace the following code for queue: User inputs the following: (5 pts)

9 13 7 48 132 73 0 76

```
int main()
{
    Queue<int> Q;
    while (i != 0)
    {
        cin >> i;
        if (i < 35)
            Q.Enqueue(i);
    }

    while (!Q.IsEmpty())
    {
        Q.Dequeue(i);
        cout << i << " ";
    }
}
```

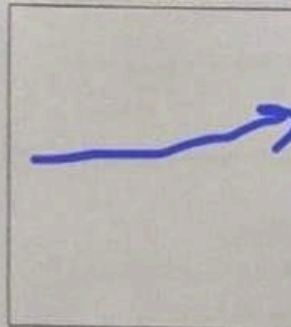
The output is: 9 13 7

```
T del ( T num )
{
    // delete from the beginning
    node <T> *q, *r;
    q = head;
    if ( q->data == num )
    {
        head = q->link;
        delete q;
        retr = q;
        while ( q != NULL )
        {
            if ( q->data == num )
            {
                r->link = q->link;
                delete q;
                return 0;
            }
            r = q;
            q = q->link;
        }
        cout << "\nElement " << num << " not Found";
        return 0;
    }
}
```

Q5: Write C++ statements for each of the following (11 pts)

1- Write the body of del function in the following ordered linklist

```
template <class T>
class orderedlinklist
{
private:
    template <class T>
    struct node {
        T data;
        node <T> *link;
    };
    node <T> *head;
public:
    orderedlinklist();
    void insert( T num );
    T del( T num );
    void display();
    T count();
}
```

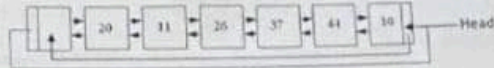


2- Given the following stack (array implementation), containing the numbers 4, 3, 6, 8 and 9. Write a method that display the contents of the stack. Items must be separated by semicolon

|   |
|---|
|   |
| 9 |
| 8 |
| 6 |
| 3 |
| 4 |

```
template <class T>
void stack <T> ::print_stack()
{
    if (isEmpty())
        cout << "Stack is empty.\n";
    else
        for (int i=0; i<3; i++)
            cout << "printing: " << array[tos--] << "\n";
}
```

- Use the following doubly link list for (7,8):



7. How The value of executing the instruction: head->next->next->data will be ?

- a) 20    ~~b) 11~~    c) 37    d) nothing

8. How The value of executing the instruction: head->next->next->prev->data will be ?

- a) 44    b) 26    c) 37    d) nothing

9. The \_\_\_\_\_ automatically executes whenever a class object goes out of scope.

- a. Constructor    c. pointer  
b. Destructor    d. exception

10. To check if a node in a BST is an internal node

- a. if(node->left == NULL || node->right == NULL)  
b. if(node->left == NULL && node->right == NULL)  
c. if(node->left != NULL && node->right != NULL)  
d. if(node->left != NULL || node->right != NULL)

Q2: TRUE/FALSE (5 pts)

TRUE  
FALSE  
FALSE  
FALSE  
FALSE

1. A stack is a First In First Out data structure.
2. A queue can only be implemented as a linked list.
3. A constructor method is used to initialize the class.
4. In this code `random[i] = 1 + rand() % 10;` all the values in array will be ( from 0 to 9 ) **F**
5. Every binary tree is a binary search tree.

Q3: For the following BST (9 pts):

a. What is the pre-order traversal for it?

1-2-3-14-7-10-11-40-30

b. What is the in-order traversal for it?

14-2-1-3-11-10-7-30-40

c. What is the post-order traversal for it?

1-3-2-7-10-40-30-11-14

